

Auditing Model Substitution in LLM APIs

Why Software Tests Fail and Hardware Attestation Wins



Trust gap in commercial LLM APIs: are you getting what you pay for?

Evidence + figures from arxiv.org/abs/2504.04715

Economic incentives drive covert swaps

Formalization (hypothesis testing)

$H_0: P_{\text{actual}} = P_{\text{spec}}$

$H_1: P_{\text{actual}} \neq P_{\text{spec}}$
Given samples from an API, decide whether responses come from the advertised distribution $P_{\text{spec}}(y|x)$ or an alternative $P_{\text{actual}}(y|x)$.



Why this matters

Black-box APIs create a verifiability gap: pricing is tied to a named model, but the delivered $P_{\text{actual}}(y|x)$ is unobservable without a trusted mechanism.



Smaller model

- e.g., serve 8B instead of 70B
- Lower GPU cost, higher margin



Quantized variant

- FP8/INT8 instead of FP16
- Often minimal benchmark drop



Fine-tuned / updated

- silent version drift

Auditing Model
Substitution in LLM
APIs

FP8 quantization preserves most accuracy — making substitution hard to detect

 Original vs. FP8 scores (mean $\pm$ error) from the source brief

Model family	MMLU (orig $\rightarrow$ FP8)	GSM8K (orig $\rightarrow$ FP8)	MATH (orig $\rightarrow$ FP8)	GPQA  (orig $\rightarrow$ FP8)
Llama-3-8B-Instruct	62.69 $\rightarrow$ 62.43 (-0.26)	61.14 $\rightarrow$ 60.90 (-0.24)	20.65 $\rightarrow$ 14.91 (-5.74)	22.62 $\rightarrow$ 20.14 (-2.48)
Llama-3-70B-Instruct	78.05 $\rightarrow$ 77.88 (-0.17)	88.06 $\rightarrow$ 87.35 (-0.71)	35.69 $\rightarrow$ 35.75 (+0.06)	29.60 $\rightarrow$ 33.12 (+3.52)
Gemma-2-9b-it	71.86 $\rightarrow$ 71.92 (+0.06)	81.80 $\rightarrow$ 79.41 (-2.39)	33.41 $\rightarrow$ 32.53 (-0.88)	28.64 $\rightarrow$ 27.81 (-0.83)
Qwen2-72B-Instruct	82.18 $\rightarrow$ 81.98 (-0.20)	86.72 $\rightarrow$ 86.82 (+0.10)	37.39 $\rightarrow$ 37.67 (+0.28)	29.93 $\rightarrow$ 31.08 (+1.15)
Mistral-7B-Instruct-v0.3	59.15 $\rightarrow$ 58.77 (-0.38)	35.90 $\rightarrow$ 32.20 (-3.70)	8.94 $\rightarrow$ 7.68 (-1.26)	21.60 $\rightarrow$ 22.72 (+1.12)

 Table: FP8 vs. original benchmark scores (source brief)

Auditing Model
Substitution in LLM

• Interpretation: For most models, FP8 scores fall within original error bars (esp. MMLU, GSM8K) — so output-APIs

Higher temperature amplifies performance gaps but increases noise

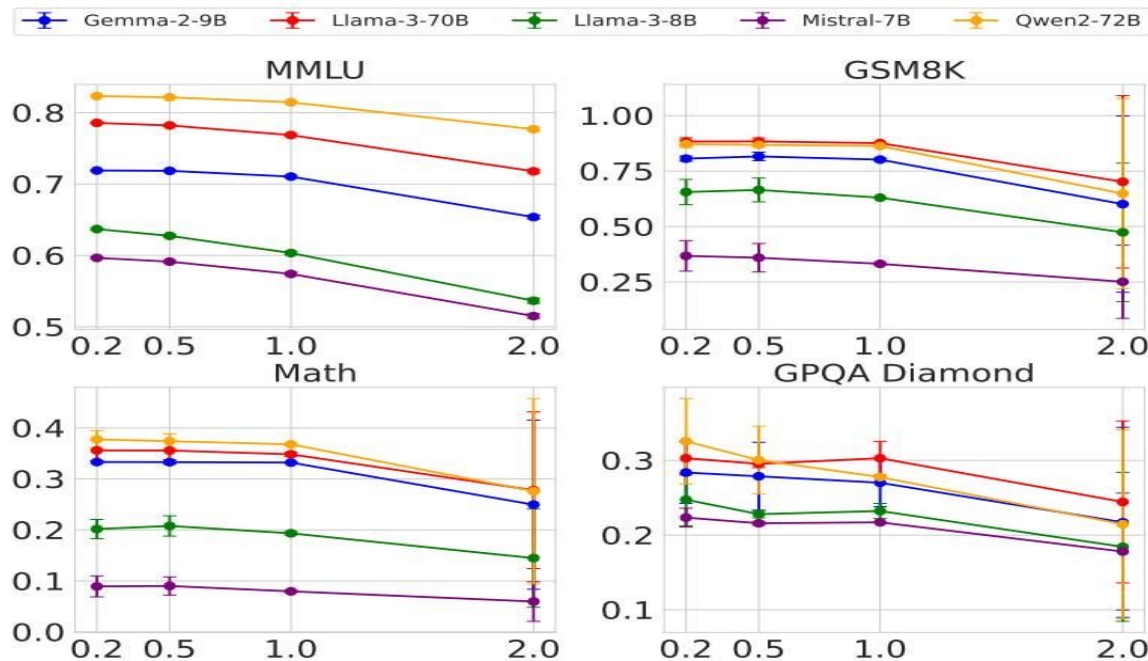


Figure 1: Benchmark accuracy vs. temperature (0.2, 0.5, 1.0, 2.0)



Empirical effect

- Substitution signal can become weak
- Variance increases → weak statistical power



Why auditing gets harder

- Observed gaps may be dominated by sampling noise.
- Tuning temperature

Log probability fingerprinting fails due to inference nondeterminism

Model	BERT Acc	T5 Acc	GPT2 Acc	LLM2Vec Acc
Llama3-70B-Instruct-FP8	50.55	50.10	50.45	49.90
Llama3-70B-Instruct-INT8	51.60	49.90	51.30	50.25
Gemma2-9b-it-FP8	49.95	50.30	51.20	49.80
Gemma2-9b-it-INT8	49.00	49.70	51.65	49.55
Mistral-7b-v3-Instruct-FP8	50.55	49.75	48.70	48.75
Mistral-7b-v3-Instruct-INT8	49.70	50.75	50.50	51.15
Qwen2-72B-Instruct-FP8	50.05	50.25	48.75	49.80
Qwen2-72B-Instruct-INT8	50.75	50.55	49.45	50.20

Figure 5: Same model, different log probs across software + hardware

- Framework differences: Transformers vs vLLM.
- Version drift: 4.40→4.50; 0.6.2→0.8.2.
- Hardware: Nondeterminism is confounded with substitution.
- Metadata-based fingerprints are not reliable in Auditing Model substitution in LLM APIs

Randomized routing at low rates makes statistical detection query-prohibitive



Attack model

Provider routes each query:

**With prob $(1-p)$:
serve advertised model**

**With prob p :
serve cheaper substitute**

Mixed response distribution

$$P_{\text{mix}} = (1-p) \cdot P_{\text{spec}} + p \cdot P_{\text{sub}}$$

As $p \rightarrow 0$, P_{mix} converges to P_{spec} :
distinguishing requires far more queries
(statistical power collapses).



Operational reality

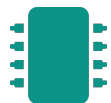
- Low-rate substitution can evade audits while still saving cost.
- Audit queries become impractically large (query-prohibitive).

Software-only auditing methods are fundamentally unreliable

Method type	Weakness	Practical limitation
Text-based statistical tests (e.g., MMD)	Need many samples; subtle substitutions indistinguishable	Query budgets become impractical; randomized substitution collapses power
Log-probability fingerprinting	Production inference nondeterminism confounds fingerprints	Different frameworks/versions/hardware yield different log-probs (false alarms)
Adversarial countermeasures	Provider can adapt to evade tests	Benchmark evasion + randomized routing defeat software-only audits



Trusted Execution Environments provide provable cryptographic model integrity



Attested inference flow

- 1) Load specific model weights
- 2) Run verified inference code
- 3) Produce attestation report
- 4) Client verifies before trust

TEEs: cryptographic attestation for verifiable inference (source brief)



Why TEEs win

- Provable: cryptographic attestation of code + weights.
- Robust: not fooled by output randomness.
- Practical: deployable today with modest overhead.



Deployment path

- TEE-backed confidential computing on modern GPUs (e.g., NVIDIA H100).
- Provider proves correct model is executed.
- Avoids ZKP infeasibility for large models.

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Key Takeaways

Close the verification gap with hardware attestation



Model substitution is economically motivated and hard to detect via black-box outputs.



FP8 quantization barely changes benchmark scores—often within error bars.



Inference nondeterminism defeats log-probability fingerprinting in production.



Randomized substitution and benchmark evasion make software tests query-prohibitive.

Call to action: treat verifiable inference as critical infrastructure.